

BAAZ JKBOSE



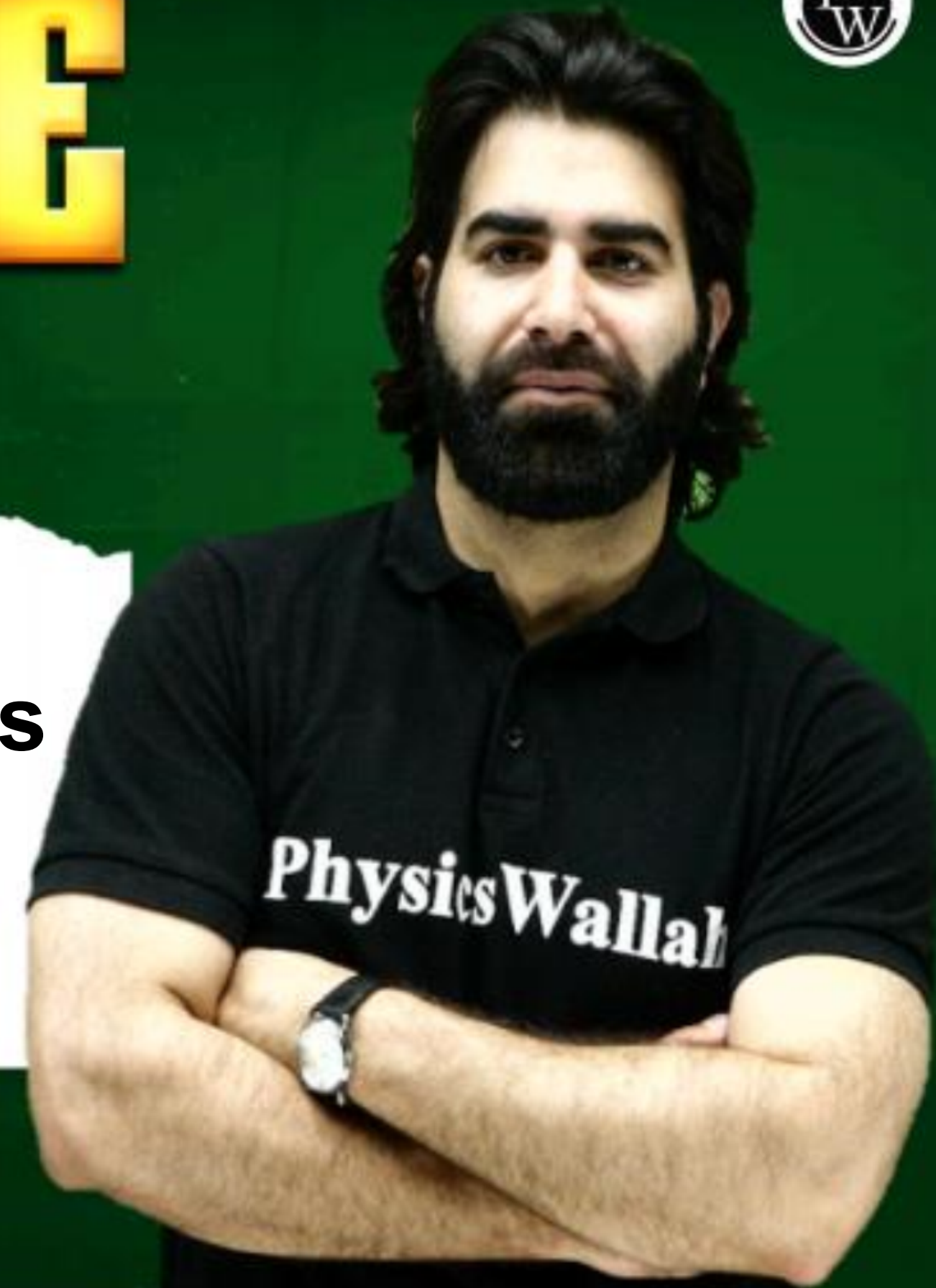
2026 CLASS-11TH

Breathing and Exchange of Gases

LECTURE-2

Zoology

Dr Adil Altaf sir

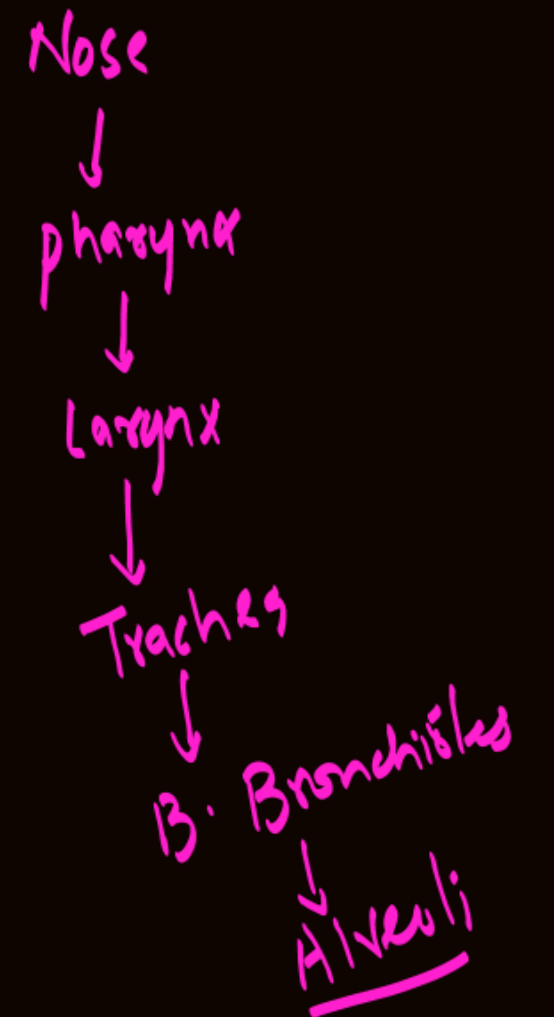


Topics To Be Covered

- 1 Human Respiratory System

Revision

- Breathing - Exchange of gases (surrounding x organism)
- Respiration - [Food $\rightarrow$ Energy]
 - Part $\rightarrow$ Breathing
- Human R.S - Nose
 - Nasal Cavity
 - Moist.
 - Filter.
 - Smell.
 - Warm.



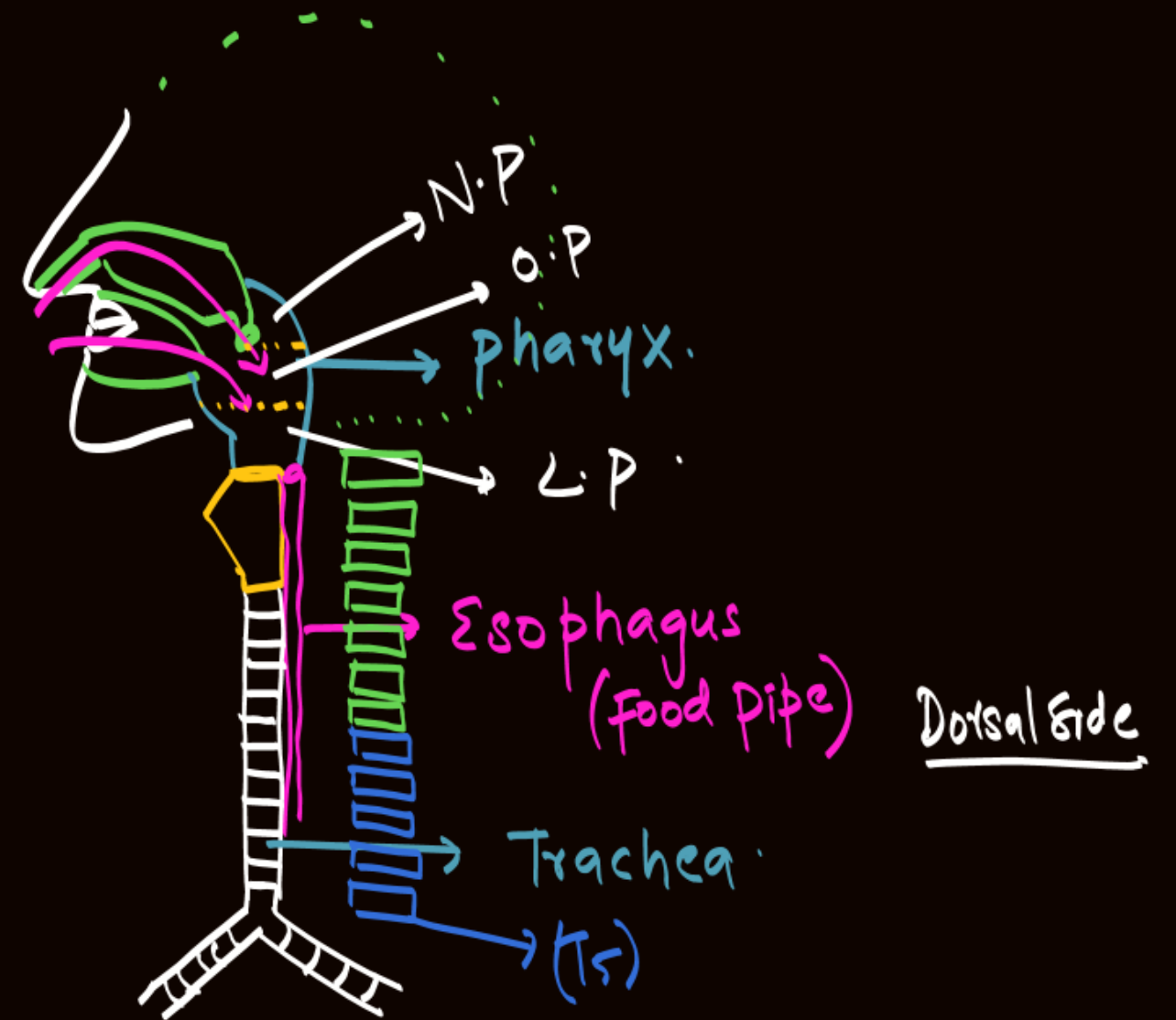
② Pharynx.

- It is a common passage for food & Air.
- It receives oral cavity and Nasal cavity.
- It has three regions

- Nasopharynx.
- Oropharynx
- Laryngopharynx.

- Various Tonsils are present here,
forming a Ring - Lymphatic Ring

Ventral Side



Larynx. (Sound Box)

↳ It is a cartilagenous organ.

↳ It is Made of Nine Cartilages.

3 Unpaired , 3 paired. (3x2)

✓ → Thyroid C
(Adam's Apple)

→ Cricoid C

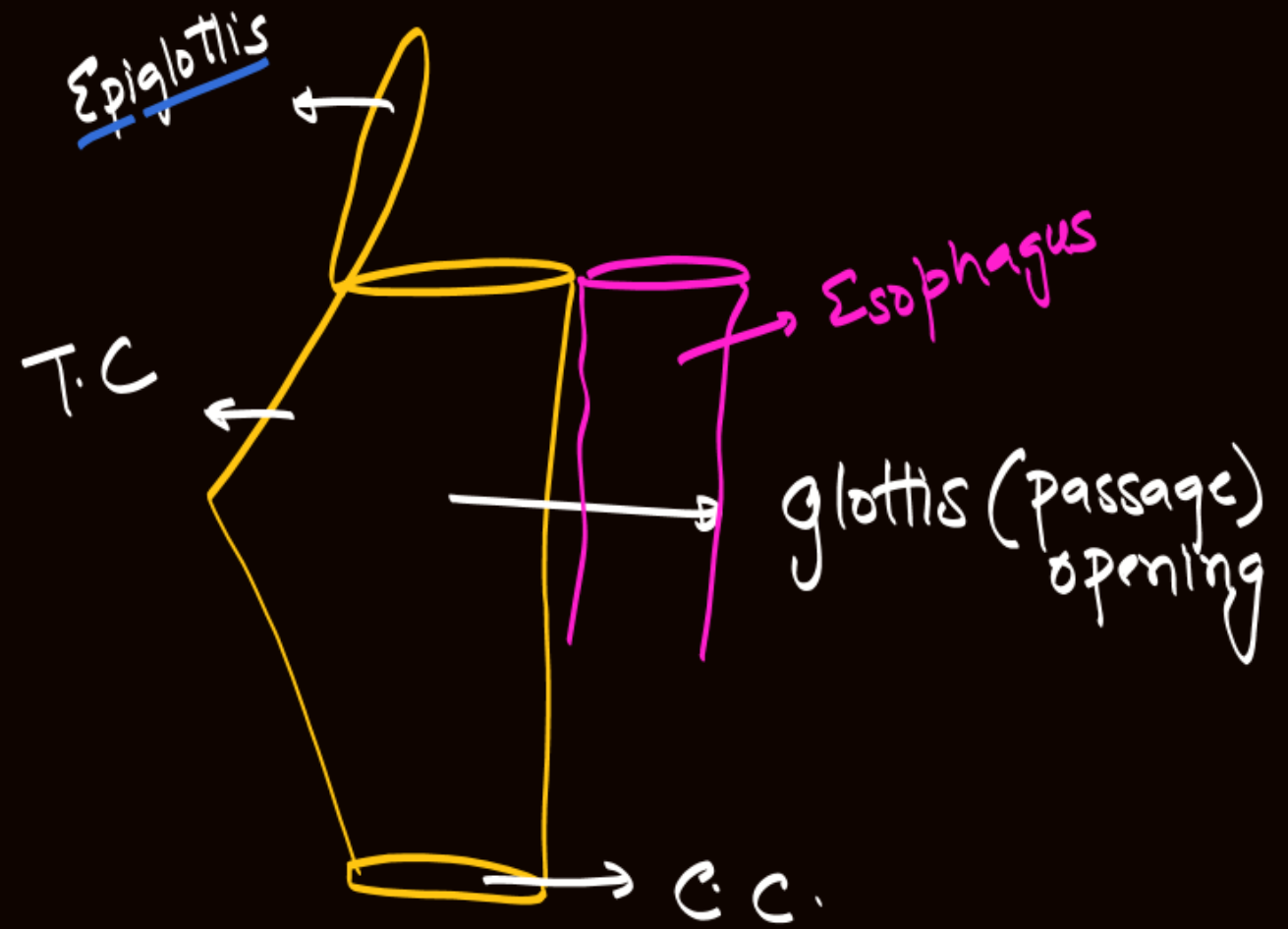
✓ → Epiglottis

Covers glottis During
Swallowing.

- Arytenoid x 2

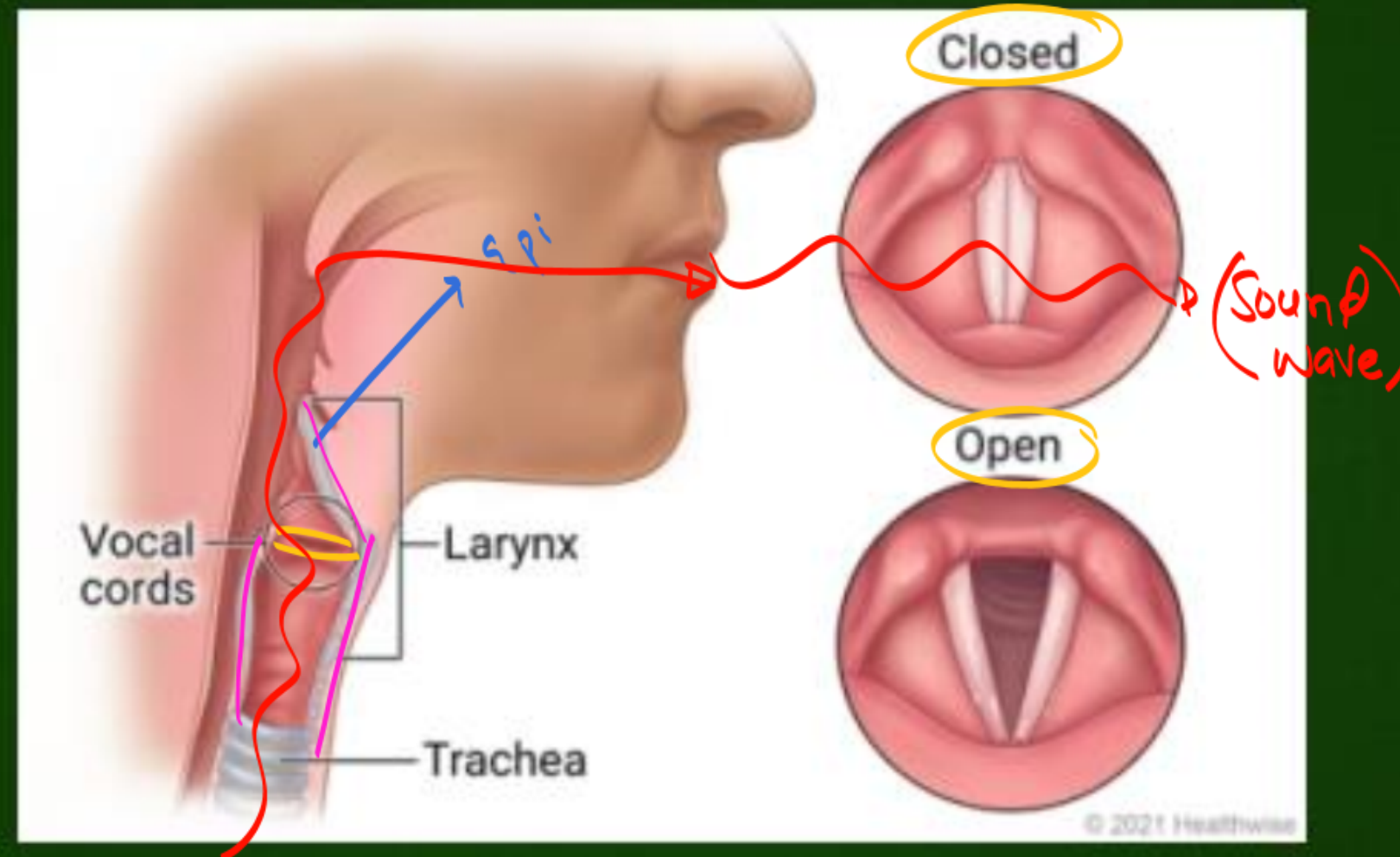
- Cuneiform x 2

- Corniculate x 2



→ Inner Side Larynx have Vocal cords

- Two upper (False)
- Two lower (True) → speech help.



Trachea.

- Also known as Wind Pipe

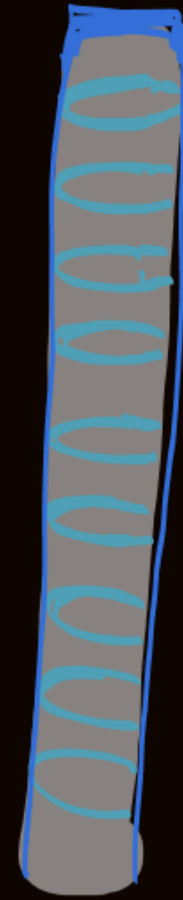
- It is a tube 10-12 cm Long.

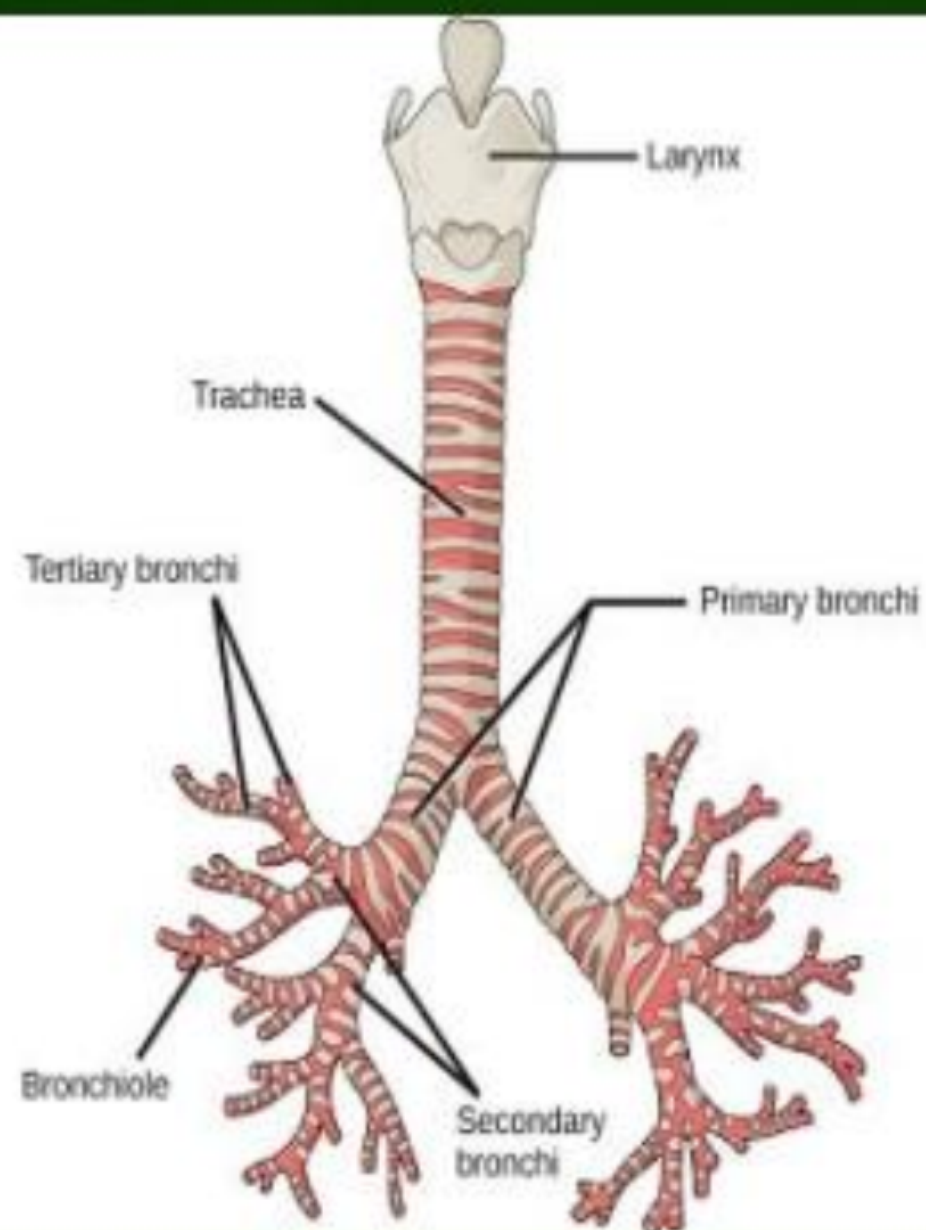
Made of C- Shaped Cartilagenous Rings.

- These Rings are incomplete , on Dorsal Side.

- These Rings make wall of Cartilage Non-Collapsible.

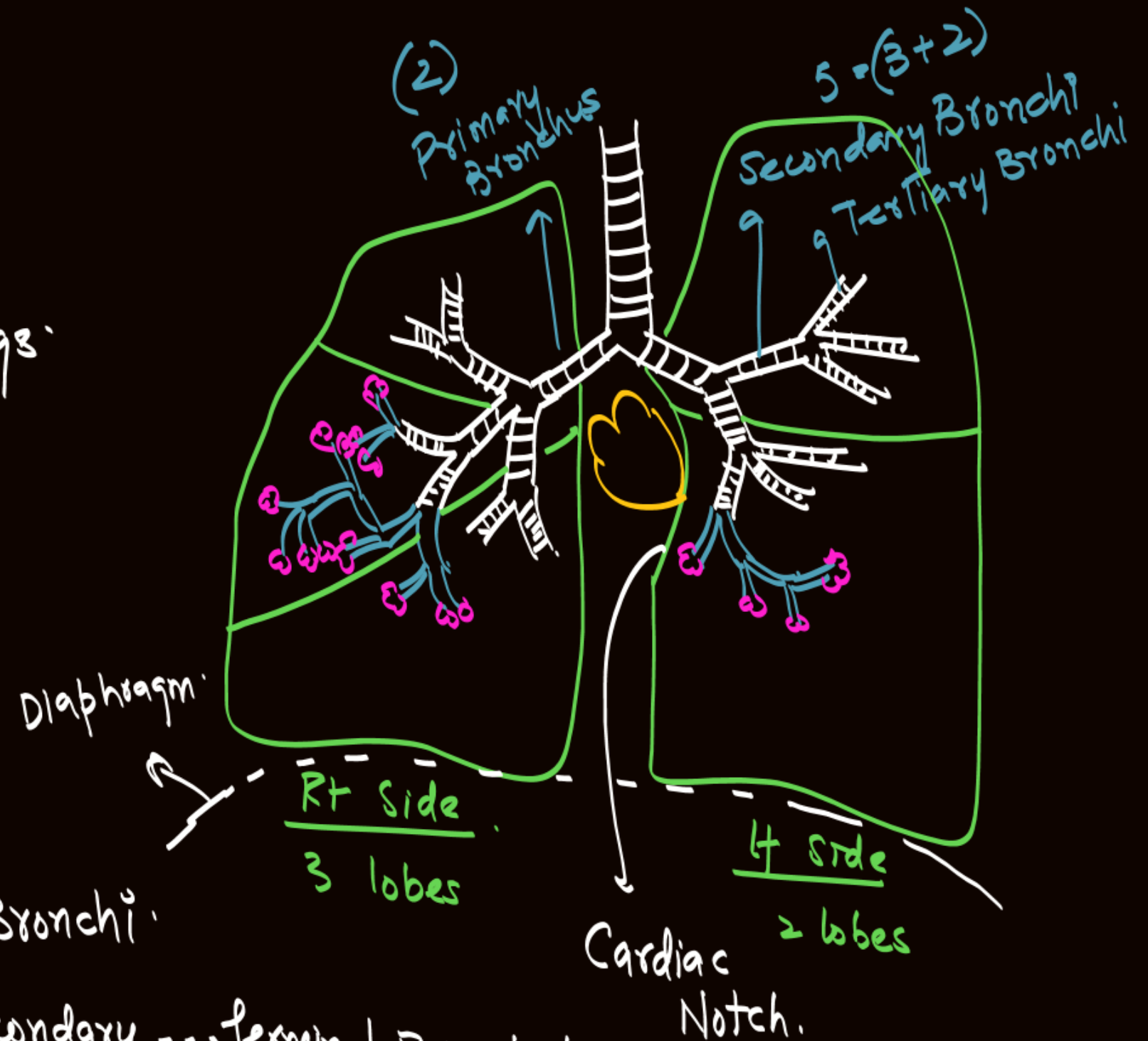
- At 5th Thoracic vertebra, Trachea bifurcates into two Bronchi





- Bronchi

- Singular - Bronchus.
- These arise from Trachea.
- These also have C-shaped cartilage Rings.
- They further divide into
 - Secondary Bronchi
 - Tertiary Bronchi



- Bronchioles

- These are smaller and Arise from tertiary Bronchi.
- They are further divided into — Primary, secondary, —, terminal Bronchioles.
- Till terminal Bronchioles, they have C-shaped Rings.

Alveoli

- Singular - Alveolus.

- These are sac like clusters.
- they are used for gaseous exchange
- they are Attached to Respiratory Bronchiole.

- Alveolus has A. Duct & A. Sac

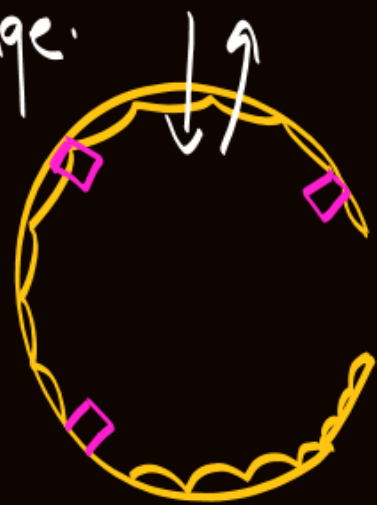
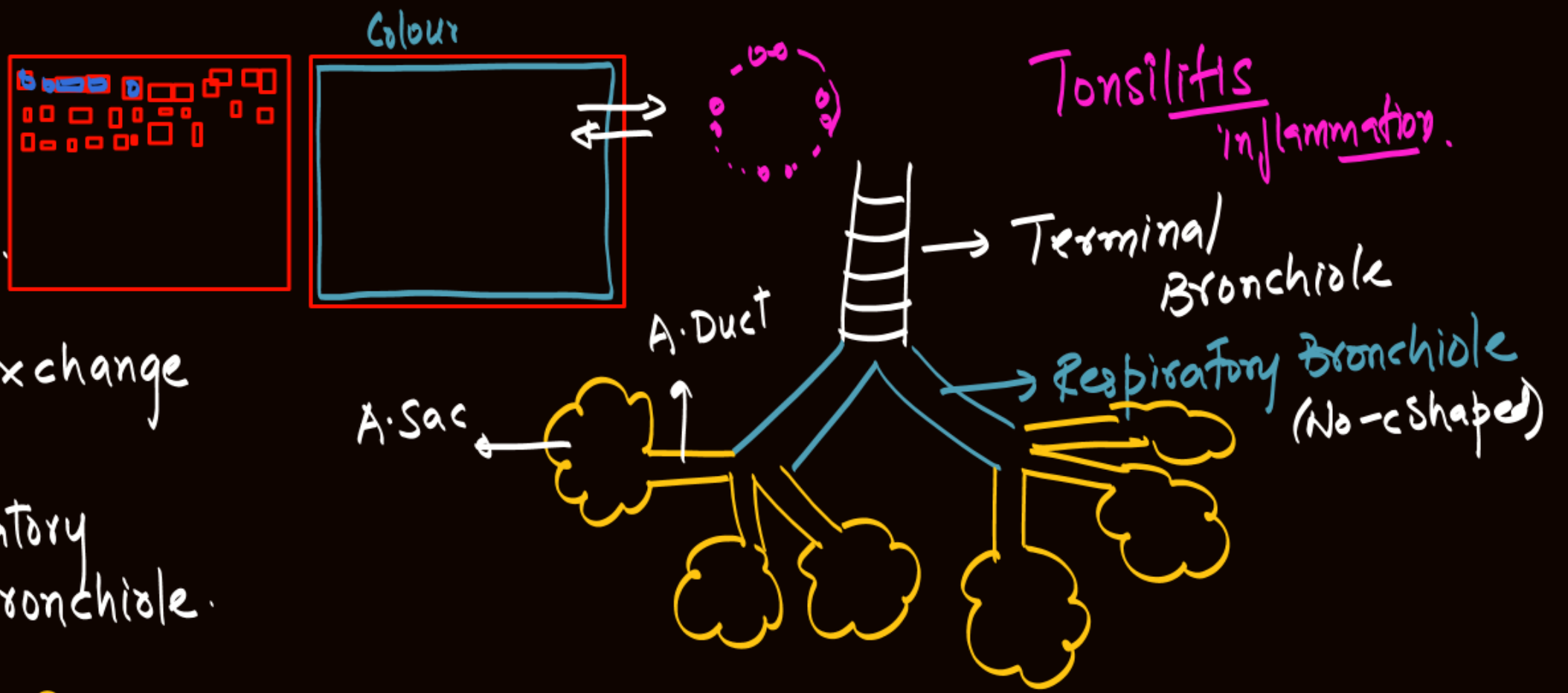
- Alveoli increase area for gaseous exchange.

→ Wall of Alveoli is Made of

95% ① Type I (pneumocytes).
- Simple Squamous Epithelium

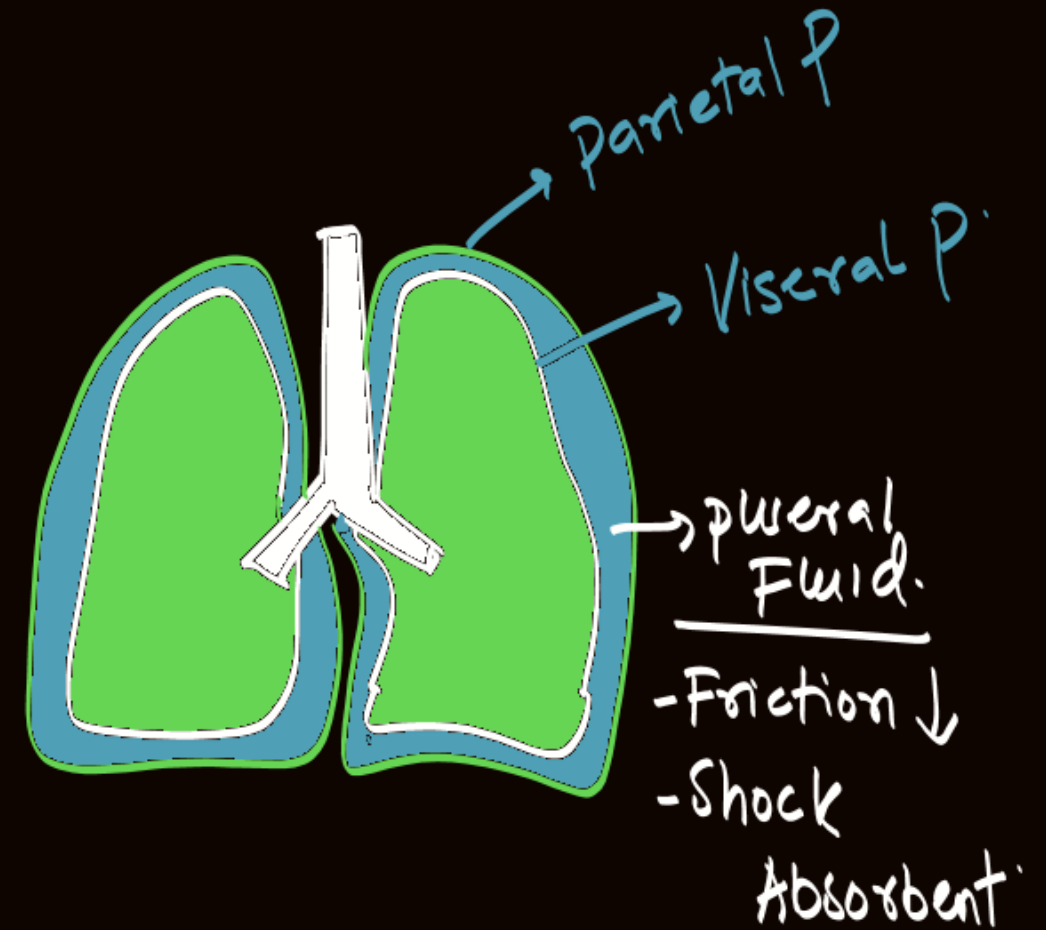
5% ② Type II (pneumocyte).

- Simple cuboidal. → produce surfactant (H/W)



Lungs

- Humans have a pair of lung.
- They contain Millions of Alveoli.
- Right lung has 3 lobes and Left Lung has Two lobes.
- Right Lung is Larger, Left is smaller but longer.
- Left lung has Cardiac Notch - to Accommodate Heart.
- Lungs are located in Thoracic Cavity.
- Covering of lungs is called Pleura → (2)
 - Inner visceral pleural
 - outer parietal pleural.
- These covering have space - pleural space. (pleural fluid)



QUESTIONS

Trachea divides into two right and left primary bronchi at the level of:

- 1 Seventh cervical vertebra
- 2 Third thoracic vertebra
- 3 Fifth thoracic vertebra
- 4 Seventh thoracic vertebra

QUESTIONS

Which of the following is entirely made of cartilage?

- 1 Nasal septum
- 2 Larynx
- 3 Glottis
- 4 Trachea

HOMework



Function of Surfactant.

Thank
You